

## ★ File System

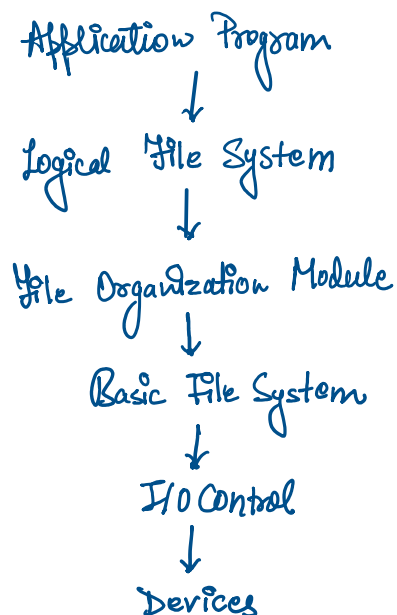
A file system is a method an OS uses to store, manage & retrieve files & directories on a storage device.

- FAT (File Allocation Table) (old version, older versions of Windows)
- NTFS (New Technology File System) (modern, Windows)
- EXT (Extended File system) (Linux, Unix)
- HFS (Hierarchical File System) (MacOS)
- APFS (Apple File System) (Introduced by Apple for their Macs & iOS devices)

Attributes: Name, Type, Size, Creation Date, Author, Last Modified, Protection

Operations: Create, Open, Read, Write, Append, Truncate, Delete

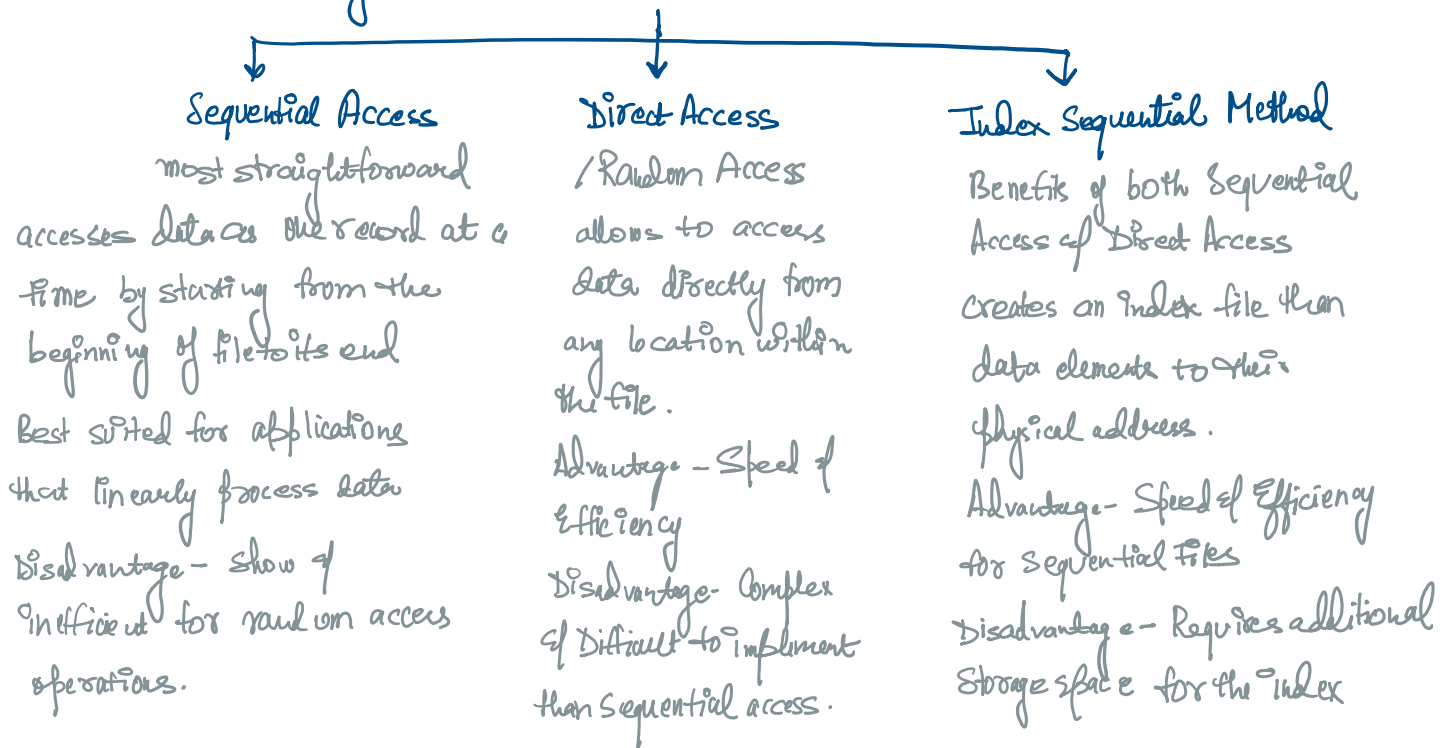
## ★ File System Structure



## ★ File Access Methods

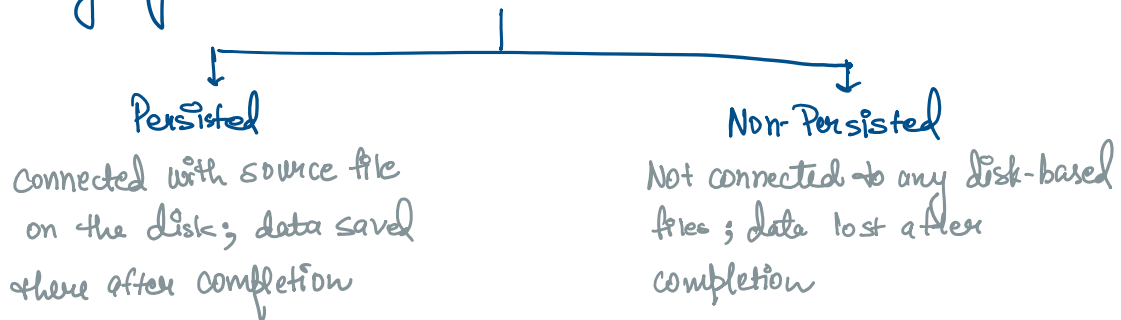
- Techniques & processes used to read from & write to files stored on storage devices.
- These methods determine how data is organized, retrieved & modified within a file system.

within a file system.



## \* Memory Mapping

Technique that allows a part of the virtual address space to be associated with a file logically.



## \* Directory Operations

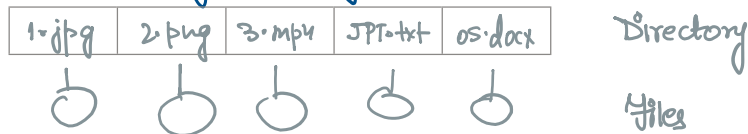
- Search for a file
- Create a file
- Delete a file
- Rename a file
- Listing of files
- File System Traversal

## \* Directory Structures

### ① Single Level Directory

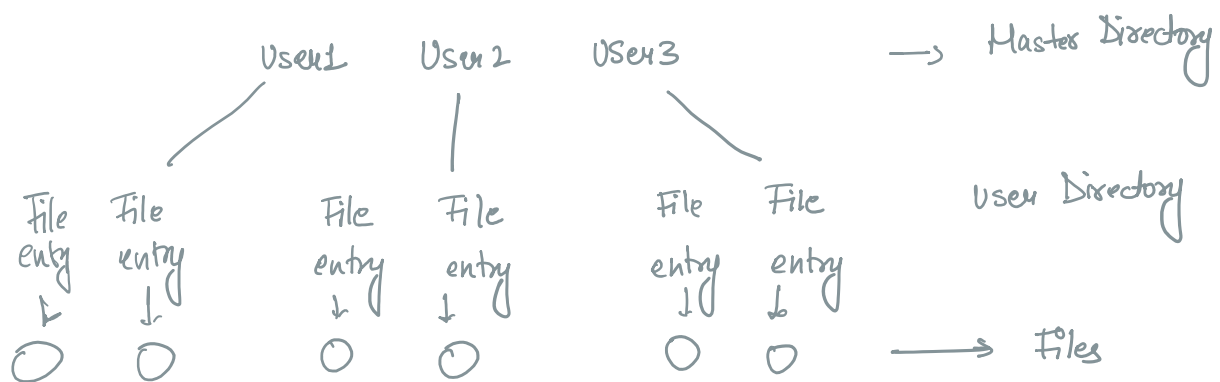
## ① Single Level Directory

- Single, flat list of all files on a storage drive
- becomes restrictive when system supports multiple users
- Can be used when system has limited no. of files of users but it lacks any method to organize or group files.

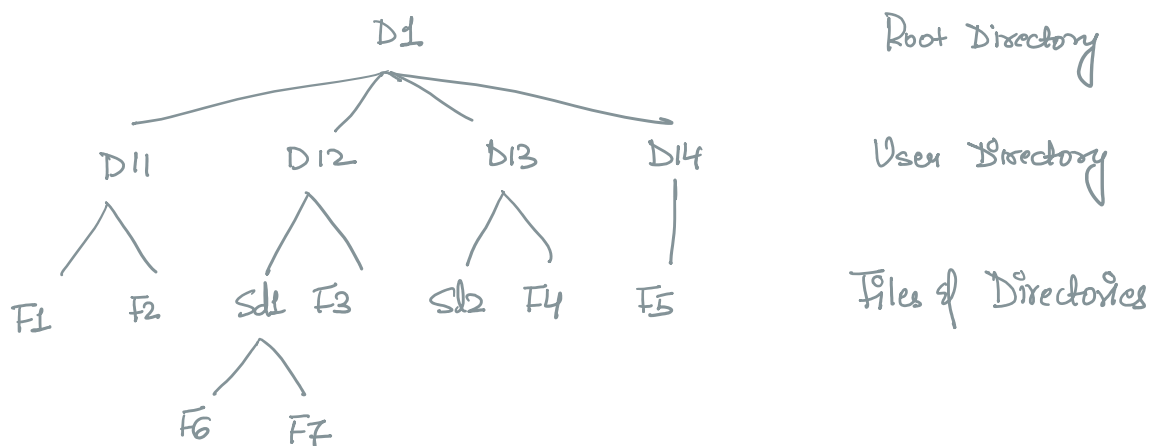


## ② Two-Level Directory

- Allows for separate directory for each user.
- same file name can exist across different directories
- first level: Master directory that contains individual users directories
- no way to group files within a user's directory



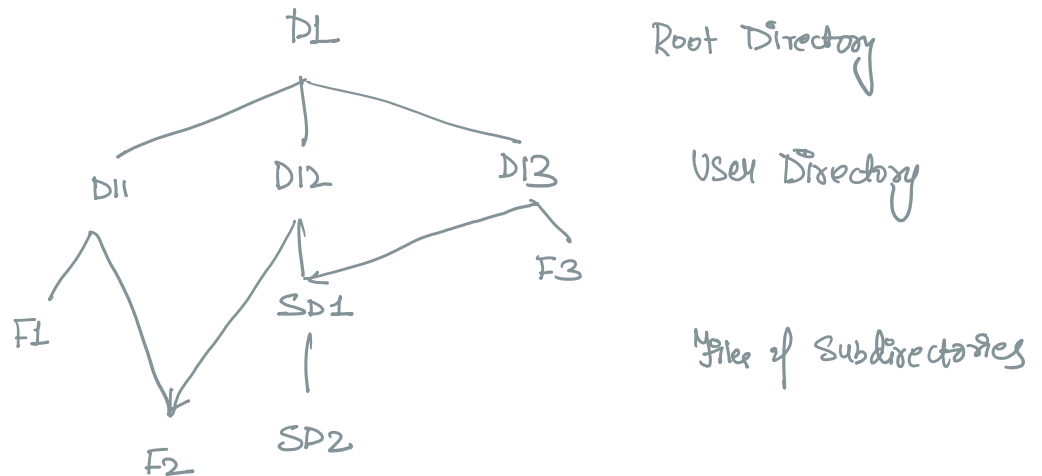
## ③ Tree Structure (Hierarchical Structure)



- searching is more efficient

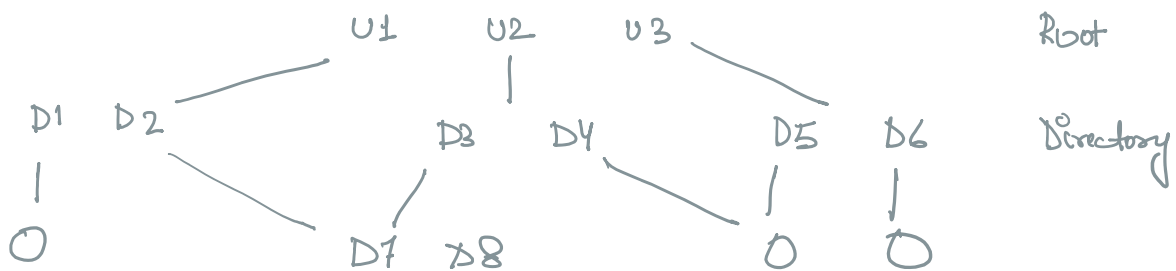
### ⑦ Acyclic

- Tree model forbids the existence of the same file in several directories.
- Links or aliases can be used to create this type of directory graph.
- Symbolic/Soft Link: Specifies a file path
- Hard Link: Actual Link to the same file
- Actual file is deleted once all the references to it have been deleted.



### ⑧ General Graph Directory Structure

- issue is figuring out how much space the files & folders have used up.



- Numerous directories can be derived from more than one parent directory in a general graph directory structure

### ★ File System Mounting

- A process in which the OS adds the directories of files from a storage device to the user's computer file system.